CS 214: Data Structures
Introduction

Slide contents follow

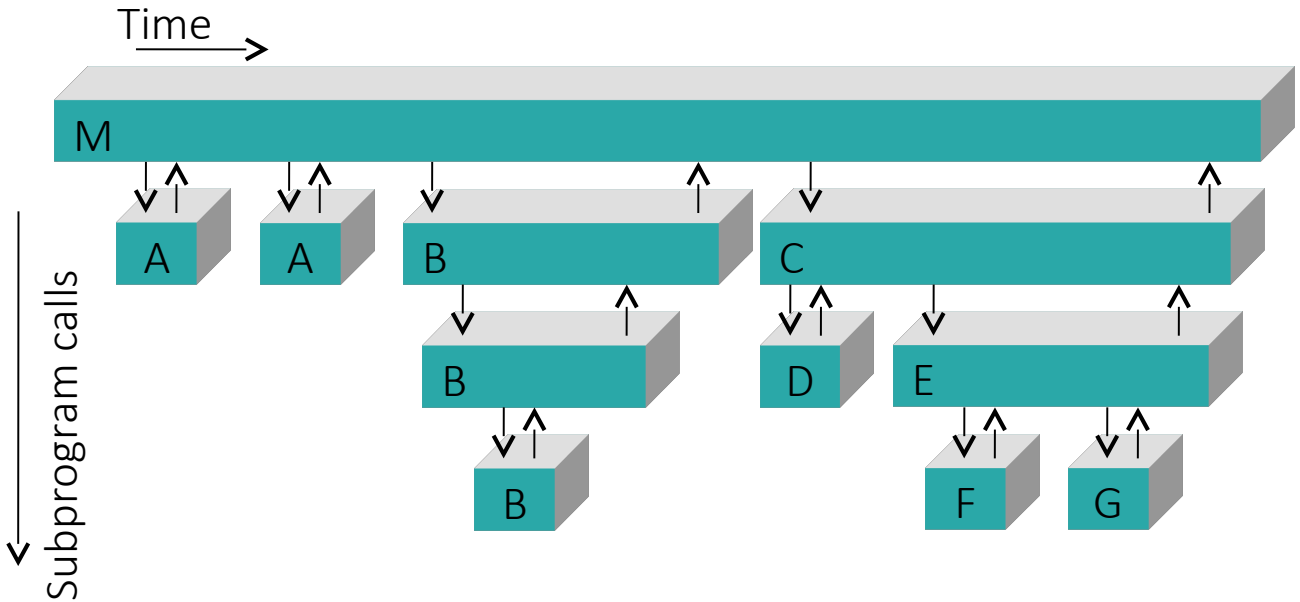
Kruse and Leung "Data Structures & Program Design in C"

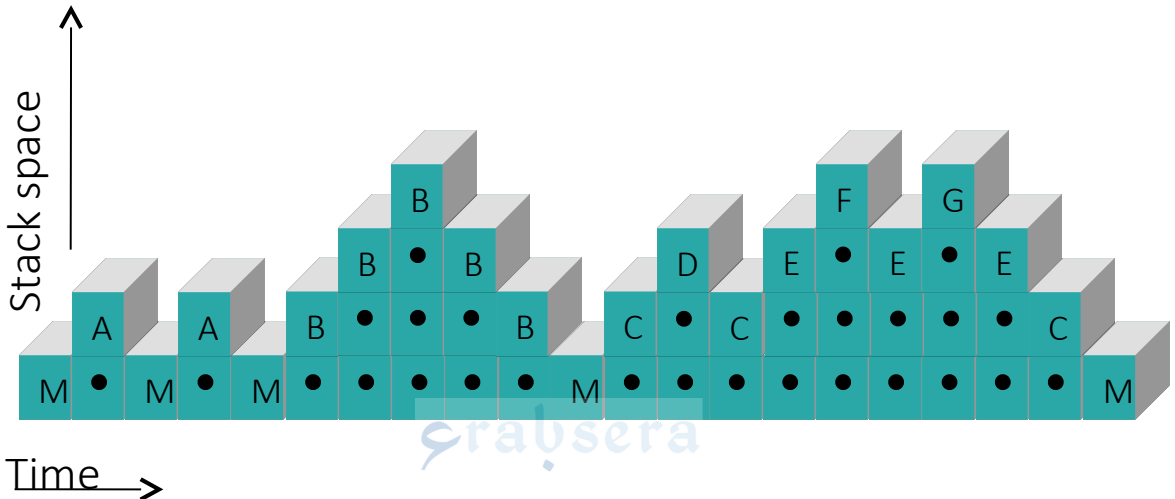
Prepared by:

Waleed A. Yousef, Ph.D.

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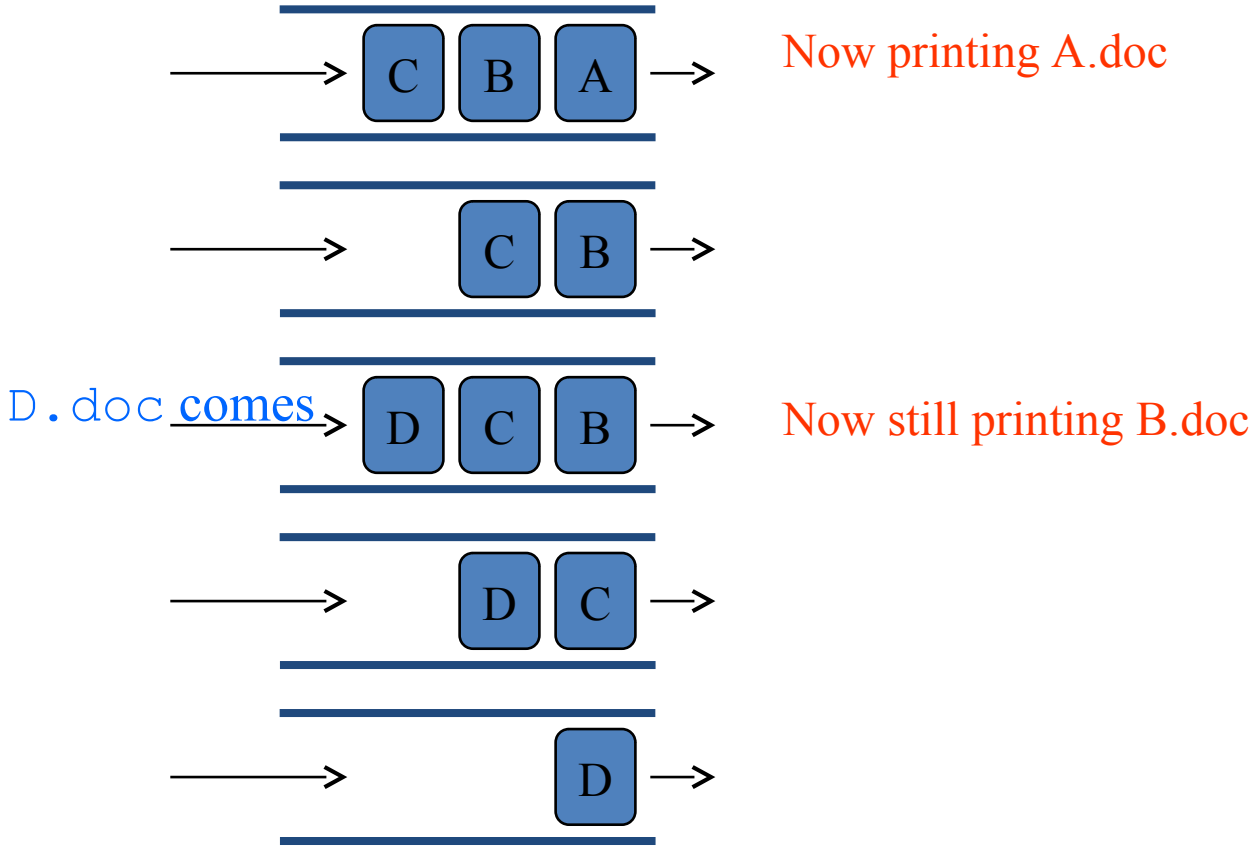
Why Data Structures?





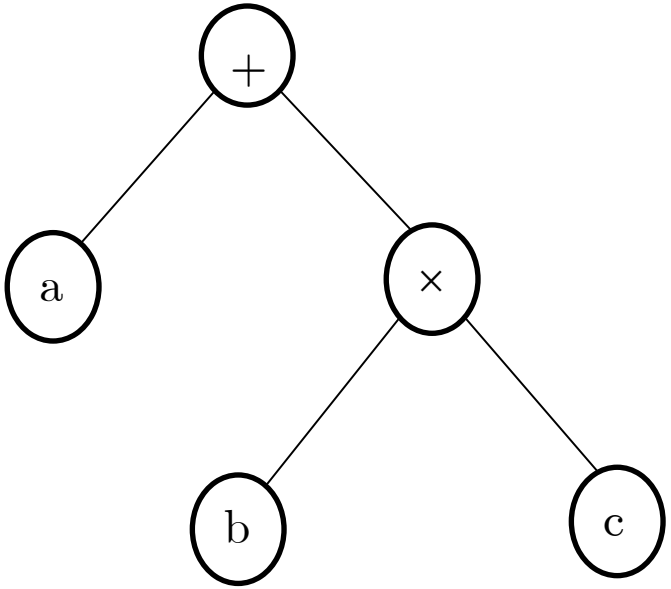
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Why Data Structures?



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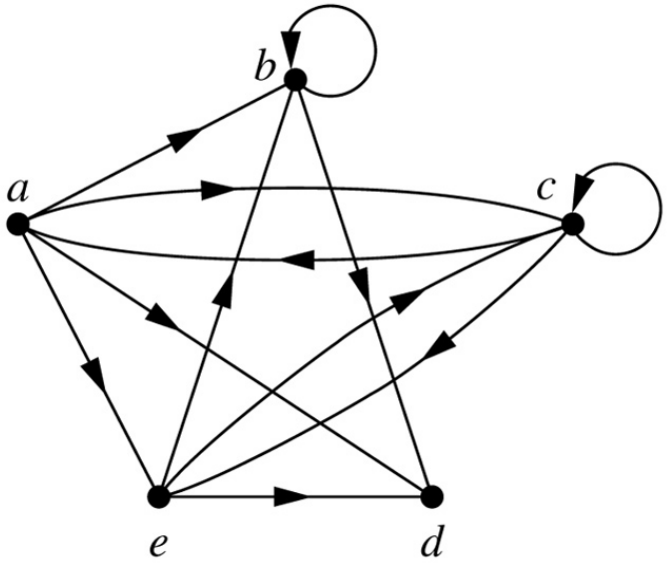
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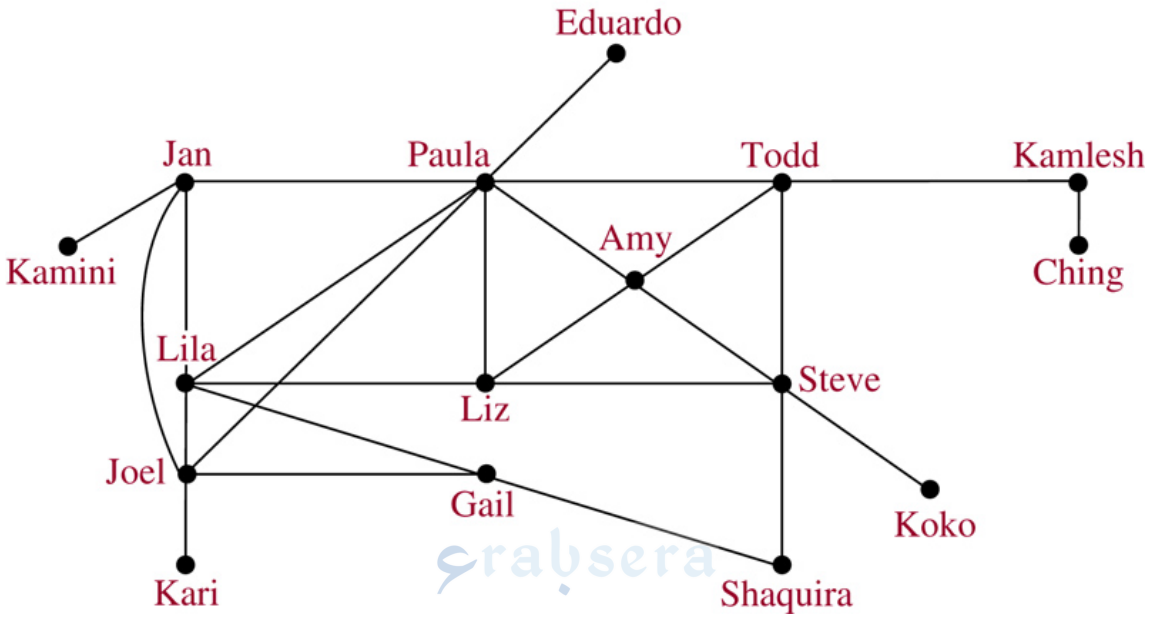


$$a + b \times c$$

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Why Data Structures?





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Course Outlines

1. Stacks.
2. Queues.
3. General Lists.

} array-based and linked based implementations

4. Search Algorithms in Lists
   a. Linear Search.
   b. Binary Search.
   c. Mathematical Analysis of Linear and Binary Search.
5. Binary Search Trees.
6. Introduction to Graphs.
7. Appendix: Introduction to Mathematical Analysis of Algorithms

- This is a “Data Structures” course, not “Data Structures and Algorithms”
- Why “Analysis of Algorithms” in a “Data Structures” course?
- Why Data Structures in C and not in pseudo-code or any other language?
- Videos and Lectures notes are self-contained (having the book is complementary)

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Warming up:
Shifting from "Hello World"-like programs to structured programs

Guidelines for choosing variable names:

1. Give special care to the choice of names for functions, and constants.
2. Use common prefixes or suffixes to associate names of the same general category.
3. Avoid deliberate misspellings and meaningless suffixes to obtain different names.
5. Avoid choosing names that are close to each other in spelling or otherwise easy to confuse
6. Be careful in the use of the letter "l" (small ell), "O" (capital oh") and "0" (zero).

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Warming up:
Shifting gears from "Hello World" to structured programs

Documentation Guidelines:

1. Place a prologue at the beginning of each function with
   a. Identification (programmer's name, date, version).
   b. Statement of the purpose of the function and method used.
   c. Pre- and Post conditions.
2. Comment on the use of variables.
3. Introduce each significant part of the program with a statement of purpose.
4. Indicate the end of each significant section.
5. Avoid comments that parrot what code does or that are meaningless jargon.
6. Explain any statement that employs a trick or whose meaning is unclear. Better still, avoid such statements.
7. The code itself should explain how the program works. The documentation should explain why it works and what it does.
8. Be sure to modify the documentation along with the program.

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